Lecture 15, 16 Notes

Wavepackets, Phase Velocity, Group Velocity.

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1. Introduction: Waves for Massive Particles

In today's lecture, we would like to understand what kind of waves represent massive particles such as electrons. In the double-slit experiment, when we derived the expression for interference, we wrote down an expression representing the light wave, which looked like

$$A\cos(kx - \omega t).$$

More generally, from your high school memory, a travelling wave is written as a function of the form

$$f(kx - \omega t),$$

where f could be any periodic or localized function. Our goal, first of all, is to ask: what type of waves might serve as a good description of massive particles like electrons? We will start from a more general wavefunction, which will later turn out to be the correct wave that describes a massive free particle like the electrons that are moving towards the slits in the double-slit experiment with electrons. It is

$$A\exp(i(kx - \omega t)).$$

First of all, how do we visualize a plane wave of this type? We could plot either its real part or its imaginary part. Let us, for simplicity, plot the real part of the wavefunction,

$$\psi(x,t) = A\cos(kx - \omega t).$$

Since it depends on both position and time, we can only plot it for a fixed instant of time.

Let us first plot it at $t = 0$. The result is a simple sinusoidal curve. Now, if we plot it again at $t = 1$, we obtain another snapshot of a similar sinusoidal shape, but displaced along the x-axis. By taking such snapshots at successive times and playing them sequentially like frames in a movie, we can visualize how the wave propagates through space and time (Click below to see the simulation).

Interactive Simulation

This is an excellent way to build intuition about travelling waves and how their form depends on both k and ω.

Now, what is the velocity of this wave? How do we describe the velocity of a wave? We can take one point on the plot, say, the peak of the cosine function at some location x and at some time t. The argument of the cosine function, which we denote by ϕ, is

$$\phi = kx - \omega t.$$

At a later instant, this peak moves to the right, as seen in the simulation. Suppose that after a small time interval Δt, the peak has moved by a distance Δx. At the new position and time, the phase of the cosine at the peak should again have the same value ϕ, so that

$$\phi = k(x + \Delta x) - \omega(t + \Delta t).$$

Subtracting the two expressions for ϕ, we obtain

$$k\Delta x - \omega\Delta t = 0,$$

or equivalently,

$$\frac{\Delta x}{\Delta t} = \frac{\omega}{k}.$$

This quantity,

$$\boxed{v_p = \frac{\omega}{k}},$$

is the velocity with which a point of constant phase (for instance, a peak) moves. It is therefore called the **phase velocity** of the wave.

You are already familiar with this concept from light waves, for which the dispersion relation is $\omega = ck$. In that case,

$$v_p = \frac{\omega}{k} = c,$$

the speed of light.

So now we know how to describe a plane wave and how to calculate its velocity. However, plane waves are not really good candidates to describe a localized particle. A plane wave has a single, well-defined wave number k and therefore a fixed wavelength $\lambda = 2\pi/k$. From de Broglie's relation,

$$p = \frac{h}{\lambda} = \hbar k,$$

we see that a plane wave represents a state with a definite momentum.

According to the uncertainty principle, a state with well-defined momentum must have a large uncertainty in position. Indeed, look at the plots we have been discussing: a plane wave is a periodic wave that extends from $x = -\infty$ to $x = +\infty$. If we recall the second postulate of quantum mechanics, $|\psi(x)|^2$ gives the probability density to find the particle at position x. For a plane wave, this probability is uniform throughout all space, meaning that we have no idea where the particle actually is.

Therefore, we must do better. We need to construct more localized waves from these extended plane waves.

2. Towards Localized Waves: The Idea of Beats

One way to construct a more localized wave is to superpose several plane waves with slightly different wave numbers. The basic idea is that, although each individual plane wave extends throughout all space, their superposition can produce a wave whose amplitude is concentrated in a limited region.

You have already encountered a closely related idea in Class 12 physics in the form of *beats*. Let us briefly revisit this familiar phenomenon to build some intuition about wave superposition.

Consider two oscillations of equal amplitude A, but with slightly different angular frequencies ω_1 and ω_2:

$$y_1(t) = A\cos(\omega_1 t), \tag{1}$$
$$y_2(t) = A\cos(\omega_2 t). \tag{2}$$

According to the principle of superposition, the resultant displacement is the sum of the two

oscillations:

$$y(t) = A\cos(\omega_1 t) + A\cos(\omega_2 t). \tag{3}$$

Using the trigonometric identity

$$\cos\alpha + \cos\beta = 2\cos\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right),$$

we obtain

$$\boxed{y(t) = 2A\cos\left(\frac{\omega_1+\omega_2}{2}t\right)\cos\left(\frac{\omega_1-\omega_2}{2}t\right).} \tag{4}$$

This expression describes a rapidly oscillating wave whose amplitude varies slowly with time when the two frequencies are close to each other. The first cosine oscillates at the average angular frequency

$$\omega_{\text{avg}} = \frac{\omega_1+\omega_2}{2},$$

while the second cosine acts as a slowly varying amplitude factor.

To see this more clearly, let us consider two sound waves with frequencies

$$f_1 = 500\,\text{Hz}, \qquad f_2 = 502\,\text{Hz}.$$

Their angular frequencies are

$$\omega_1 = 2\pi(500), \tag{5}$$
$$\omega_2 = 2\pi(502). \tag{6}$$

Consequently,

$$\frac{\omega_1+\omega_2}{2} = 2\pi(501), \qquad \frac{|\omega_1-\omega_2|}{2} = 2\pi(1).$$

The resultant displacement therefore becomes

$$y(t) = 2A\cos(2\pi(501)t)\cos(2\pi t). \tag{7}$$

The wave oscillates rapidly at an average frequency of 501 Hz, while its amplitude is modulated by the slowly varying factor

$$2A\cos(2\pi t).$$

The amplitude envelope repeats after one second. The sound consequently becomes loud and soft periodically, producing the phenomenon known as *beats*.

The number of beats heard per second is called the beat frequency. In general,

$$\boxed{f_{\text{beat}} = |f_1 - f_2|.} \tag{8}$$

For our example, the beat frequency is 2 Hz.

You can explore how two waves of slightly different frequencies combine to produce beats by clicking on the following simulation:

Interactive Simulation: Wave Interference and Beat Frequency

Try changing the two frequencies and observe how the rate of the amplitude modulation changes.

What does this teach us about localized waves? The important lesson is that superposition can produce a resultant wave with a structure that is not apparent from either component wave individually. In the example above, two waves of nearly equal frequencies produce a slowly varying amplitude envelope.

However, beats alone do not produce a truly localized wave packet: the two component waves still extend throughout space. To construct a wave packet localized in position, we must superpose plane waves with different wave numbers and choose their amplitudes and phases appropriately. We will now develop this idea more systematically.

3. Group Velocity: Motion of the Wave Envelope

In the previous section, we saw that the superposition of two waves with slightly different frequencies produces beats. We now restore the spatial dependence of the waves and ask a more interesting question: how does the amplitude modulation propagate through space?

Consider two plane waves of equal amplitude A, with slightly different angular frequencies and wave numbers:

$$y_1(x,t) = A\cos(\omega t - kx), \tag{9}$$
$$y_2(x,t) = A\cos\big[(\omega+\Delta\omega)t - (k+\Delta k)x\big]. \tag{10}$$

Here, $\Delta\omega$ and Δk are small compared with ω and k, respectively.

According to the principle of superposition, the resultant wave is

$$y(x,t) = y_1(x,t) + y_2(x,t). \tag{11}$$

Using the trigonometric identity

$$\cos\alpha + \cos\beta = 2\cos\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right),$$

we obtain

$$y(x,t) = 2A\cos\left[\left(\omega+\frac{\Delta\omega}{2}\right)t - \left(k+\frac{\Delta k}{2}\right)x\right] \times \cos\left(\frac{\Delta\omega}{2}t - \frac{\Delta k}{2}x\right). \tag{12}$$

Since $\Delta\omega \ll \omega$ and $\Delta k \ll k$, we can approximate the first cosine by $\cos(\omega t - kx)$. Thus,

$$\boxed{y(x,t) \approx 2A\cos(\omega t - kx)\cos\left(\frac{\Delta\omega}{2}t - \frac{\Delta k}{2}x\right).} \tag{13}$$

This expression is particularly interesting. It describes a rapidly oscillating wave whose amplitude varies slowly with both position and time.

The first cosine represents the rapid oscillations, while the second cosine determines the slowly varying amplitude modulation. The latter produces a sequence of wave groups, or an envelope, that propagates through space.

3.1. Group Velocity

Let us now determine the velocity with which the amplitude envelope moves. From Eq. (13), the slowly varying factor is

$$\cos\left(\frac{\Delta\omega}{2}t - \frac{\Delta k}{2}x\right).$$

Consider a particular point on this envelope, corresponding to a fixed value of its phase,

$$\Phi = \frac{\Delta\omega}{2}t - \frac{\Delta k}{2}x.$$

As the envelope propagates, suppose this point moves from x to $x + \Delta x$ during a time interval Δt. Since we are following the same point on the envelope, its phase must remain unchanged. Therefore,

$$\Phi = \frac{\Delta\omega}{2}(t + \Delta t) - \frac{\Delta k}{2}(x + \Delta x).$$

Subtracting the two expressions for Φ, we obtain

$$\frac{\Delta\omega}{2}\Delta t - \frac{\Delta k}{2}\Delta x = 0.$$

Rearranging,

$$\frac{\Delta x}{\Delta t} = \frac{\Delta\omega}{\Delta k}.$$

The velocity of the envelope is called the **group velocity**, denoted by v_g. Hence,

$$\boxed{v_g = \frac{\Delta\omega}{\Delta k}.} \tag{14}$$

When we consider a continuous range of wave numbers rather than just two nearby values, we take the limit $\Delta k \to 0$. The group velocity is then given by

$$\boxed{v_g = \frac{d\omega}{dk}.} \tag{15}$$

This is the general expression for the group velocity in terms of the dispersion relation $\omega = \omega(k)$.

The phase velocity tells us how rapidly the individual oscillations move, while the group velocity tells us how rapidly the envelope moves.

For a localized particle, the wave packet is constructed by superposing plane waves with a range of wave numbers. The motion of this packet is therefore naturally associated with the group velocity.

4. A Finite Range of Wave Numbers

As you can see from the simulation, this superposition gives a wave that is somewhat localized, certainly more so than a single plane wave, but still extends infinitely in space because it is composed of only a discrete sum of components. A truly localized wave requires a continuous superposition of plane waves with a range of wave numbers. We therefore define a general wave packet as

$$\psi(x,t) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}\phi(k)e^{i(kx-\omega(k)t)}\,dk. \tag{16}$$

Here, $\phi(k)$ specifies the amplitude of the component wave with wave number k; this is known as the Fourier transform of the wave function. This gives wave packets.

Let us now look at a simple example to see how moving from a discrete sum to an integral over wave numbers produces a wave that is more localized in space.

We begin with the Fourier-type representation introduced above, but to keep things simple we will take its real part. Consider a range of wave numbers centered about some k_0, extending from $k_0 - \Delta k/2$ to $k_0 + \Delta k/2$. Within this interval, let us assume that the amplitude $\phi(k)$ is a constant a_0, and zero elsewhere. Then the real part of the wave function at $t = 0$ may be written as

$$y(x) = a_0 \int_{k_0-\Delta k/2}^{k_0+\Delta k/2} \cos(kx)\, dk. \tag{17}$$

Evaluating the integral gives

$$y(x) = \frac{a_0}{x}\left[\sin\left(\left(k_0 + \frac{\Delta k}{2}\right)x\right) - \sin\left(\left(k_0 - \frac{\Delta k}{2}\right)x\right)\right]. \tag{18}$$

Using the trigonometric identity

$$\sin(A + B) - \sin(A - B) = 2\cos A \sin B, \tag{19}$$

we obtain

$$\boxed{y(x) = \frac{2a_0}{x}\cos(k_0 x)\sin\left(\frac{\Delta k\, x}{2}\right).} \tag{20}$$

This is the wave packet at $t = 0$. At later instants, it moves to the right, which can be calculated by the above procedure. The above result shows two distinct factors:

- The term
$$\frac{\sin(\Delta k\, x/2)}{x}$$
describes a slowly varying envelope that determines the extent of localization in space.
- The $\cos(k_0 x)$ factor represents the rapid oscillations within the envelope.

The prefactor $1/x$ in the expression ensures that the amplitude of the wave decays for large values of $|x|$; in other words, the wave packet dies off as we move far from the central region. This spatial damping makes the function normalizable (we will see what that means) and physically meaningful, in contrast to the infinite plane wave.

5. Examples

A. Application: Free Nonrelativistic Particle

For a free nonrelativistic particle,

$$\omega(k) = \frac{\hbar k^2}{2m}, \tag{21}$$

so

$$\omega'(k) = \frac{\hbar k}{m}. \tag{22}$$

Hence, for a packet peaked at k_0,

$$v_g = \frac{\hbar k_0}{m} = \frac{p_0}{m}, \tag{23}$$

which equals the classical particle velocity, while

$$v_p = \frac{\omega_0}{k_0} = \frac{\hbar k_0}{2m} = \frac{v_g}{2}. \tag{24}$$

Thus, the phase velocity is half the classical particle velocity.

B. Relativistic Dispersion Relation

For a free relativistic particle, the energy and momentum are related by

$$E^2 = p^2c^2 + m^2c^4. \tag{25}$$

Using the de Broglie relations $E = \hbar\omega$ and $p = \hbar k$, we obtain the relativistic dispersion relation

$$\boxed{\omega(k) = \sqrt{(ck)^2 + \left(\frac{mc^2}{\hbar}\right)^2}.} \tag{26}$$

The phase velocity is

$$v_p = \frac{\omega}{k} = \frac{E}{p}, \tag{27}$$

and the group velocity is

$$v_g = \frac{d\omega}{dk} = \frac{dE}{dp} = \frac{pc^2}{E}. \tag{28}$$

Multiplying these two gives a simple and important identity:

$$\boxed{v_p v_g = c^2.} \tag{29}$$

This means that for any relativistic particle,

- The group velocity v_g equals the actual particle velocity v.
- The phase velocity
$$v_p = \frac{c^2}{v_g} = \frac{c^2}{v}$$
is always greater than c for a massive particle ($v < c$).